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RF POWER SPLITTING AND CONTROL

Abstract

Embodiments disclosed herein include an apparatus that includes a radio frequency (RF) power supply, and a power splitter electrically coupled to the RF power supply. In an embodiment, the apparatus includes a plurality of impedance matches electrically coupled to the power splitter, where the power splitter is configured to split RF power from the RF power supply in order to supply the RF power to each of the plurality of impedance matches. In an embodiment, the apparatus further includes a plurality of chambers, wherein each chamber is electrically coupled to a different one of the plurality of impedance matches.

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Background/Summary

BACKGROUND

1) Field

[0001] Embodiments relate to the field of semiconductor manufacturing and, in particular, to RF power splitting and control.

2) Description of Related Art

[0002] Plasma processing tools are used throughout the manufacturing process of semiconductor devices. For example, plasma processing tools may be used in deposition process, etching process, and/or treatment processes. A common form of plasma processing tool is a radio frequency (RF) plasma processing tool. In an RF plasma processing tool, a power generator is provided to generate the RF power. The RF power is delivered to a chamber so that the RF power can be coupled to processing gasses in order to ignite and/or sustain a plasma within the chamber.

[0003] Impedance matching is used to provide more efficient operation. For example, an impedance match is often provided between the power generator and the chamber in order to match the plasma load impedance to the characteristic impedance of the system. The impedance match may be a variable impedance match in order to account for changing plasma load impedances within the chamber (as a result of, for example, process variable changes during a process recipe).

[0004] In a one-to-one system (i.e., a system with a single power generator that is coupled to a single chamber), the control of the variable impedance match is relatively straight forward. However, as multiple chambers are coupled to a single power generator, the control of the system becomes more complex.

SUMMARY

[0005] Embodiments disclosed herein include an apparatus that includes a radio frequency (RF) power supply, and a power splitter electrically coupled to the RF power supply. In an embodiment, the apparatus includes a plurality of impedance matches electrically coupled to the power splitter, where the power splitter is configured to split RF power from the RF power supply in order to supply the RF power to each of the plurality of impedance matches. In an embodiment, the apparatus further includes a plurality of chambers, wherein each chamber is electrically coupled to a different one of the plurality of impedance matches.

[0006] Embodiments disclosed herein may also include an apparatus with a radio frequency (RF) power generator and a power splitter coupled to the RF power generator. In an embodiment, the apparatus further includes a plurality of impedance matches, where each of the plurality of impedance matches is coupled to the RF power generator, and where the power splitter is configured to split RF power from the RF power generator to supply RF power to each of the plurality of impedance matches. In an embodiment, each of the plurality of impedance matches includes an actuator configured to alter an impedance of a respective impedance match. In an embodiment, the apparatus further includes a plurality of reflected power sensors, where each of the plurality of reflected power sensors is electrically coupled between the power splitter and a different one of the plurality of impedance matches. In an embodiment, the apparatus further includes a controller, where the controller is configured to receive an input signal from each of the plurality of reflected power sensors and use the input signal to drive the actuators in order to alter the impedance of the respective impedance match.

[0007] Embodiments may also include an apparatus for controlling a plurality of impedance matches that includes coupling circuitry and analog control circuitry configured to receive incident and reflected radio frequency (RF) voltages and RF current signals from a plurality of sensors, and an RF control platform. In an embodiment, the RF control platform includes programmable logic, a real time processing unit (RTPU), a message controller, and a processor. In an embodiment, the apparatus further includes a plurality of motor control interfaces, where the plurality of motor control interfaces are configured to control impedances of the plurality of impedance matches.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1A is schematic illustration of a plasma processing tool that comprises a single RF power supply that is coupled to a plurality of chambers, where the RF power is split by a power splitter and each chamber has its own variable impedance match, in accordance with an embodiment.

[0009] FIG. 1B is a schematic illustration of a plasma processing tool that comprises a single RF power supply that is coupled to a plurality of chambers for batch processing substrates, where the RF power is split by a power splitter and each chamber has its own variable impedance match, in accordance with an embodiment.

[0010] FIG. 1C is a schematic illustration of a plasma processing tool that comprises a single RF power supply that is coupled to a plurality of chambers, where the RF power is split by a power splitter and each chamber comprises a plurality of variable impedance matches, in accordance with an embodiment.

[0011] FIG. 2 is a plan view illustration of an RF plasma processing tool with an RF power distribution system that comprises a splitter and a plurality of matches downstream of the splitter, in accordance with an embodiment.

[0012] FIG. 3A is a circuit diagram of an RF power splitter, in accordance with an embodiment.

[0013] FIG. 3B is a circuit diagram of an RF power splitter with improved isolation, in accordance with an embodiment.

[0014] FIG. 4A is a circuit diagram of a pi network match that can be used in an RF power distribution system that comprises a splitter and a plurality of matches downstream of the splitter, in accordance with an embodiment.

[0015] FIG. 4B is a circuit diagram of an inverted L network match that can be used in an RF power distribution system that comprises a splitter and a plurality of matches downstream of the splitter, in accordance with an embodiment.

[0016] FIG. 4C is a circuit diagram of a two stage L network match that can be used in an RF power distribution system that comprises a splitter and a plurality of matches downstream of the splitter, in accordance with an embodiment.

[0017] FIG. 5 is a circuit diagram of an RF plasma processing tool that comprises a splitter and a plurality of impedance matches between the splitter and their respective chambers, in accordance with an embodiment.

[0018] FIG. 6 is a block diagram of a controller for operating an RF power distribution system that comprises a splitter and a plurality of matches downstream of the splitter, in accordance with an embodiment.

[0019] FIG. 7 illustrates a block diagram of an exemplary computer system that may be used in conjunction with a processing tool, in accordance with an embodiment.

DETAILED DESCRIPTION

[0020] Embodiments described herein include apparatuses and methods for RF power splitting and control. In the following description, numerous specific details are set forth in order to provide a thorough understanding of embodiments. It will be apparent to one skilled in the art that embodiments may be practiced without these specific details. In other instances, well-known aspects are not described in detail in order to not unnecessarily obscure embodiments. Furthermore, it is to be understood that the various embodiments shown in the accompanying drawings are illustrative representations and are not necessarily drawn to scale.

[0021] Various embodiments or aspects of the disclosure are described herein. In some implementations, the different embodiments are practiced separately. However, embodiments are not limited to embodiments being practiced in isolation. For example, two or more different

embodiments can be combined together in order to be practiced as a single device, process, structure, or the like. The entirety of various embodiments can be combined together in some instances. In other instances, portions of a first embodiment can be combined with portions of one or more different embodiments. For example, a portion of a first embodiment can be combined with a portion of a second embodiment, or a portion of a first embodiment can be combined with a portion of a second embodiment and a portion of a third embodiment.

[0022] The embodiments illustrated and discussed in relation to the figures included herein are provided for the purpose of explaining some of the basic principles of the disclosure. However, the scope of this disclosure covers all related, potential, and/or possible, embodiments, even those differing from the idealized and/or illustrative examples presented. This disclosure covers even those embodiments which incorporate and/or utilize modern, future, and/or as of the time of this writing unknown, components, devices, systems, etc., as replacements for the functionally equivalent, analogous, and/or similar, components, devices, systems, etc., used in the embodiments illustrated and/or discussed herein for the purpose of explanation, illustration, and example.

[0023] As noted above, coupling a single RF power distribution system with a plurality of chambers results in the need for a significantly more complex control scheme. Typically, the impedance match is provided between the RF generator and the power splitter. One significant issue with such a design is that cross-talk between chambers becomes hard to manage. For example, the individual chambers may each have different plasma load impedances. When the splitter is downstream of the match, a change in the load condition or transient will infringe on the voltage and current flow through the splitter network and create a secondary transient for impedance and power balancing.

[0024] Accordingly, embodiments disclosed herein may include an RF power distribution system that comprises a power splitter that is provided upstream of a plurality of impedance matches (e.g., at least one impedance match for each chamber). That is, the portion of the power distribution system upstream of the power splitter is a fixed impedance system. Such an architecture can provide multiple benefits.

[0025] One benefit is that voltage and current handling is improved. For a characteristic impedance, the power splitting can be optimized for transmitting lower voltages and/or currents compared to architectures where the impedance match is between the RF generator and the power splitter. Further, power distribution with an optimized characteristic impedance is significantly less complex. This yields a lower cost and enables reliable process power distribution. The RF stress on the system is also decreased.

[0026] Complexity may be reduced due to the improved isolation provided by embodiments described herein. As used herein, “isolation” may refer to the immunity of one branch of the splitting system from impairing the performance of another branch. As described above, the existing architecture has little (or no) protection for such cross-talk. In embodiments disclosed herein, the improved isolation allows for the reduction (or elimination) of secondary transients, which makes impedance matching and power balancing less complicated. For example, transient effects may be reduced by potentially orders of magnitude.

[0027] The complexity is also reduced through improvements in impedance matching control. For example, each impedance match in the RF power distribution system may include its own actuators in order to modify the impedance. In combination with a dedicated sensor for each impedance match, the control system can monitor reflected power and make adjustments to one or more variable passive devices (e.g., a capacitor) in order to improve the impedance matching along a single branch of the RF power distribution system.

[0028] Improved ability to scale systems is also enabled by embodiments disclosed herein. For example, scaling can include adding additional RF generators at different RF power levels. When adding multiple RF generators, the power splitter to multiple impedance matches architecture has superior isolation compared to existing solutions. For example, isolation may be greater than

several orders of magnitude of a second RF generator impairing the first RF generator.

[0029] RF distribution systems that have a single splitter that feeds a plurality of impedance matches may be useful in multiple application spaces. In one embodiment, distributed process chambers on a single mainframe with independent vacuum systems for single substrate processing may benefit from the use of the RF distribution systems described herein. Similarly, distributed process chambers on a single mainframe that have a common vacuum system for batch substrate processing may benefit from the use of the RF distribution systems described herein.

[0030] Referring now to FIG. 1A, a schematic illustration of an RF plasma processing tool **100** is shown, in accordance with an embodiment. In an embodiment, the RF plasma processing tool **100** may be a multi-chamber **140** tool. The RF plasma processing tool **100** may be any type of RF plasma processing tool, such as a plasma enhanced atomic layer deposition (PEALD) tool, a plasma enhanced chemical vapor deposition (PECVD) tool, a physical vapor deposition (PVD) tool, a plasma etching tool, a plasma treatment tool, or the like. In FIG. 1A, chambers **140** include four chambers (e.g., Chamber A to Chamber D). Though, it is to be appreciated that embodiments may include an RF plasma processing tool **100** with two or more chambers **140**. The chambers **140** may be part of a single mainframe or cluster tool. Though, in other embodiments, the chambers **140** may be part of separate systems.

[0031] In an embodiment, the chambers **140** may each comprise their own vacuum systems for single substrate **141** processing. The substrates **141** may include any type of substrate **141** that is processed with a plasma. For example, the substrate **141** may be a semiconductor substrate, such as a silicon wafer or the like. The substrate **141** may have a standard form factor (e.g., 300 mm wafer, 450 mm wafer, etc.). In other embodiments, substrate **141** may be a dielectric substrate, a ceramic substrate, a glass substrate, or the like. Such substrate **141** may have panel, half-panel, quarter-panel, etc. form factors which may be common for plasma processing of such substrates.

[0032] In an embodiment, an RF power distribution system may supply RF power to each of the chambers **140**. The RF power may be coupled to one or more gasses provided within the chambers **140** in order to ignite and/or sustain an RF plasma used to process the substrates **141**. In an embodiment, the RF power distribution system may include a multiple branch configuration, with each branch being coupled to one of the chambers **140**.

[0033] In an embodiment, the RF power distribution system may comprise an RF generator **110**. RF generators **110** may also be referred to as an RF power supply in other embodiments. The RF generator **110** may be configured to provide RF signals at a particular frequency and with a desired power. In some embodiments the power can be modulated. The frequency and power levels may be chosen based on the desired process outcomes within the chambers **140**. In an embodiment, the frequency may be 13.56 MHz, though other frequencies may also be used. In yet another embodiment, the frequency of the RF power generator **110** may also be variable in order to provide frequency tuning.

[0034] In the illustrated embodiment, a single RF generator **110** is shown. However, embodiments may also include multiple RF generators **110** in order to provide different RF powers into the RF power distribution system. As will be described in greater detail below, the isolation benefits of the architectures described herein may allow for improved isolation (e.g., at least several orders of magnitude) between a first RF generator **110** and a second RF generator **110**.

[0035] In an embodiment, the RF power distribution system may also comprise a power splitter **120**. The power splitter **120** may have a single input to receive RF power from the RF generator **110** (e.g., along a coaxial cable) and a plurality of outputs to feed RF power to the plurality of chambers **140**. In the illustrated embodiment, a single output line extends out of the power splitter **120**, and that single line is then branched. This illustration is done in an effort to simplify the illustration in FIG. 1A. In a more realistic view, the “splitting” of the RF power is done within the power splitter **120**, and a plurality of output lines will exit the power splitter **120**.

[0036] In an embodiment, any suitable RF power splitting architecture may be used as the power

splitter **120**. In an embodiment, the power splitter **120** may be a high isolation splitter or a low isolation splitter. The low isolation splitter may have less complexity and undergo less RF stress. The high isolation splitter may have a more complex circuitry, with the benefit of providing better isolation between branches of the RF power distribution system, compared to the low isolation splitter. A more detailed explanation of the power splitter **120** is provided in greater detail below. [0037] In an embodiment, the power splitter **120** may be configured to evenly distribute the RF power to each of the branches within the RF power distribution system. For example, all of the chambers **140** may receive substantially the same amount of RF power. In other embodiments, the power splitter **120** may be configured to provide unequal RF power distribution. In an embodiment, the unequal RF power distribution may be suitable for when chambers **140** are running different process recipes or the like. The power splitter **120** may be fixed, so that the distribution of power between each branch of the RF power distribution system is constant. In other embodiments, the power splitter **120** may have controls in order to adjust the percentage of the RF power delivered to any of the branches in the RF power distribution system. In an embodiment, the portion of the power distribution system upstream of the power splitter **120** is a fixed impedance system. As described above, the use of a fixed impedance system allows for improved voltage and current handling and reduced complexity in the system. Fixed impedances may also avoid standing waves in distribution cables, and optimizes for efficiency and power delivery. Though, other embodiments may include a variable impedance system upstream of the power splitter **120**. For example, an RF generator with a PIN diode may be provided upstream of the power splitter **120**.

[0038] In an embodiment, the RF power distribution system may further comprise a plurality of impedance matches **130**. As shown, in FIG. 1A a set of four impedance matches **130** (e.g., Match A to Match D) are provided in order to provide a one-to-one ratio of impedance matches **130** to chambers **140**. Though, as will be described in greater detail below, the number of impedance matches **130** may be different than the number of chambers **140**.

[0039] In an embodiment, each of the impedance matches **130** may be electrically coupled between the power splitter **120** and the respective chamber **140** to which the impedance match **130** is connected. The interconnects between the power splitter **120** and the impedance matches **130** may comprise coaxial cables. Similarly, the impedance matches **130** may be connected to the chambers **140** by coaxial cabling or the like. As used herein, each “branch” of the RF power distribution system may refer to the interconnect from the power splitter **120** to the impedance match **130**, the impedance match **130**, and the interconnect (if present) between the impedance match **130** and the chamber **140**. The chamber **140** may be considered as part of the branch in some instances, though inclusion of the chamber **140** as part of the branch is not necessary.

[0040] In an embodiment, the impedance matches **130** may comprise any suitable matching circuitry. For example, the impedance matches **130** may comprise a pi network match, an L network match, an inverted L network match, a two stage L network match, or the like. The impedance matches **130** may be variable impedance matches **130** in order to adjust the impedance in order to account for changes in the plasma load impedance within the chamber **140**. The impedance matches **130** may, therefore, include one or more actuators for actuating passive components (e.g., capacitors or inductors) in order to achieve the desired impedance. The impedance matches **130** may be controlled by a controller (not shown) that receives sensor data from the RF power distribution system in order to calculate the necessary changes that should be made to the impedance matches **130**.

[0041] Referring now to FIG. 1B, a schematic illustration of an RF plasma processing tool **101** is shown, in accordance with an embodiment. In an embodiment, the RF plasma processing tool **101** in FIG. 1B may be similar to the RF plasma processing tool **100** in FIG. 1A, with the exception of the chambers **140** that are coupled to the RF power distribution system. Instead of having independent vacuum systems, the chambers **140** may be distributed process chambers on a single mainframe (not shown) that have a common vacuum system. This allows for easier batch substrate

142 processing. As shown in each chamber **140**, a batch **142** (e.g., two or more substrates **141**) may be processed at the same time, or at nearly the same time.

[0042] Referring now to FIG. **1C**, a schematic illustration of an RF plasma processing tool **102** is shown, in accordance with an embodiment. In an embodiment, the RF plasma processing tool **102** in FIG. **1B** may be similar to the RF plasma processing tool **101** in FIG. **1B**, with the exception of the number of impedance matches **130** coupled to each chamber **140**. For example, each chamber **140** in FIG. **1C** is coupled to a pair of impedance matches **140**. For example, chamber **140A** is coupled to impedance match **130A1** and impedance match **130A2**, and chamber **140B** is coupled to impedance match **130B1** and impedance match **130B2**.

[0043] Accordingly, a number of branches in the RF power distribution system may be different than a number of chambers **140** in the tool **102**. In the particular embodiment shown in FIG. **1C**, the ratio of impedance matches **130** to chambers **140** is 2:1. Though, other ratios may also be used in some embodiments. In the illustrated embodiment, the chambers **140** are shown as batch **142** processing chambers **140**, similar to those shown in FIG. **1B**. In other embodiments, multiple impedance matches **130** per chamber **140** may also be used for chambers **140** for single substrate **141** processing, similar to those shown in FIG. **1A**.

[0044] Referring now to FIG. **2**, a plan view illustration of a platform tool **200** is shown, in accordance with an embodiment. The platform tool **200** may be a semiconductor processing tool such as a PEALD tool, PECVD tool, a PVD tool, a plasma etching tool, a plasma treatment tool, or the like. In an embodiment, the cluster tool may comprise an equipment front end module (EFEM) **207**. The EFEM **207** may receive front opening unified pods (FOUPs) **208** or other wafer transport devices. A wafer handling robot within the EFEM **207** transfers wafers from the FOUP **208** to a load lock **206**. The load lock **206** is coupled to a transfer chamber **205** that is held at a vacuum pressure. That is, the load lock **206** allows for the transition from an atmospheric pressure environment to a vacuum environment. A wafer handling robot within the transfer chamber **205** can distribute wafers from the load lock **206** to any of the chambers **240** that are coupled to the transfer chamber **205**. For example, ten chambers **240** are shown in FIG. **2**. Though, two or more chambers **240** may be used in some embodiments. The chambers **240** may be similar to any of the chambers described in greater detail herein.

[0045] In an embodiment, each of the chambers **240** may be supplied RF power by an RF power distribution system. The RF power distribution system may comprise an RF generator **210**. While a single RF generator **210** is shown in FIG. **2**, it is to be appreciated that multiple RF generators **210** may be included in the RF power distribution system in order to implement scaling, as described in greater detail herein. In an embodiment, a power splitter **220** is electrically coupled to the RF generator **210**. In an embodiment, the RF splitter **220** may be a high isolation power splitter **220** or a low isolation power splitter **220**.

[0046] In an embodiment, the power splitter **220** may split RF power from the RF generator **210** into a plurality of branches. In an embodiment, each branch is coupled to one of the chambers **240**. For example, in FIG. **2** the chambers **240** and the branches are provided in a one-to-one ratio. Though, chambers **240** may be coupled to more than one branch of the RF power distribution system in other embodiments.

[0047] In an embodiment, an impedance match **230** may be provided along each branch of the RF power distribution system. The impedance matches **230** may be variable matches that can be configured to alter their impedance in order to account for changing plasma load impedances within the chambers **240**. While the impedance matches **230** may provide variable impedances, the portion of the power distribution system upstream of the power splitter **220** is a fixed impedance system. As described above, the use of a fixed impedance system allows for improved voltage and current handling and reduced complexity in the system.

[0048] Referring now to FIGS. **3A** and **3B**, examples of circuit diagrams for power splitters **320** that can be used in the RF power distribution system are shown, in accordance with an

embodiment. While two specific examples are shown in FIGS. 3A and 3B, it is to be appreciated that any suitable circuit capable of splitting RF power into two or more branches may be used, in accordance with various embodiments. In the particular embodiments shown in FIGS. 3A and 3B, the power splitters **320** are examples of power splitters. The power splitter **320** in FIG. 3A represents an example of a low isolation power splitter **320**, and the power splitter **320** in FIG. 3B represents an example of a high isolation power splitter **320**. More generally, the high isolation power splitter **320** will provide a greater extend of isolation between the branches compared to the low isolation power splitter **320**.

[0049] Referring now to FIG. 3A, the power splitter **320** is fed RF power from an RF generator **310**. The RF power is then split along a plurality of lines **322** within the power splitter **320** (e.g., one line for each branch that will be used in the RF power distribution system). For example, four lines **322A-322D** are shown in the illustrated embodiment. However, embodiments may include two or more lines **322**. In an example, each line **322** includes substantially similar passive components (e.g., capacitors, inductors, and transmission lines). In the illustrated embodiment, the main path of the line **322** comprises a series of passive elements such as an inductor, while an arm extending off the main path (in a tee configuration) may comprise a capacitor that is grounded at one terminal. In the embodiment shown, each main path exits the power splitter **320** towards a branch (e.g., Branches 1-4). Each branch may comprise an impedance match and a chamber, similar to those described in greater detail herein.

[0050] While the low isolation power splitter **320** in FIG. 3A may not be optimized to provide the highest level of isolation between the branches, such a power splitter **320** may have the benefit of lower complexity and fewer components. This can reduce the overall cost of the power splitter **320**. Additionally, such a power splitter **320** may also provide the benefit of lowering RF stress within the RF power distribution system.

[0051] Referring now to FIG. 3B, a circuit diagram of a high isolation power splitter **320** is shown, in accordance with an additional embodiment. The high isolation power splitter **320** may also be fed RF power from an RF generator **310**. Similar to the embodiment in FIG. 3A, the power splitter **320** may comprise a plurality of lines **322** (e.g., two or more lines **322**). For example, lines **322A-322D** are shown in FIG. 3B.

[0052] In an embodiment, each line **322** may comprise passive components on a main path and one or more arms. In FIG. 3B, a single arm is shown, but it is to be appreciated that two or more arms may be used in some embodiments. In an embodiment, the one or more arms may be provided between passive elements on the line **322**. For example, In FIG. 3B, the arms are provided between capacitors. Though, it is to be appreciated that the arms may be provided at any suitable location along the line **322**. In an embodiment, the arm may comprise one or more passive elements such as an inductor or a capacitor, with the inductor being connected to a ground in the shown embodiment. In the embodiment shown, each main path exits the power splitter **320** towards a branch (e.g., Branches 1-4). Each branch may comprise an impedance match and a chamber, similar to those described in greater detail herein.

[0053] Additionally, pairs of the main lines may be electrically coupled together by a bridge **324**. For example, bridge **324A** couples the line **322A** to the line **322B**, and the bridge **324B** couples the line **322C** to the line **322D**. Each bridge **324** may comprise one or more resistors. In an embodiment, the bridge **324A** may be electrically coupled to the bridge **324B** by a common node **326**. In some instances, the power splitter **320** may also comprise a delta variant high isolation splitter architecture.

[0054] In an embodiment, the use of a high isolation power splitter **320** provides the benefit of decreasing the cross-talk between branches. This makes the control of the RF power distribution system easier from a controls perspective. However, the additional components and complexity may increase the cost of the high isolation power splitter **320** compared to the low isolation power splitter **320**.

[0055] In FIGS. 3A and 3B, each line 322 is considered as being identical to each other. However, in other embodiments the lines 322 may allow for different percentages of the RF power to flow to each branch. For example, changes to one or more of the component values, such as capacitance values or inductance values, may be used to alter the percentage of RF power flowing to each branch. Additionally, while fixed power splitters 320 are shown, other embodiments may include variable power splitters 320 that can be operated to change the percentage of power flown into each branch (e.g., by using variable components).

[0056] Referring now to FIGS. 4A-4C, some examples of impedance matches 430 that may be used in the RF power distribution system are shown, in accordance with an embodiment. FIG. 4A illustrates a pi network match, FIG. 4B illustrates an inverted L network match, and FIG. 4C illustrates a multi-stage L network match. While three types of impedance matches 430 are shown as suitable architectures, other types of matches may also be used. For example, an L network match, a multi-stage inverted L network match, or the like may also be used in some embodiments.

[0057] Referring now to FIG. 4A, a circuit diagram of an impedance match 430 with a pi network is shown, in accordance with an embodiment. The impedance match 430 may comprise a main line 434 and a pair of arms 433A and 433B. The main line 434 may comprise a series of inductors, and the arms 433A and 433B may each comprise passive elements such as a capacitor or an inductor that are coupled to a ground. In an embodiment, an input 431 of the impedance match 430 is fed RF power from a power splitter (not shown), and the output 432 provides RF power to a chamber (not shown). Further, in some embodiments, the match stages may be separated by transmission lines (not shown) or at least partially comprised of transmission lines.

[0058] Referring now to FIG. 4B, a circuit diagram of an impedance match 430 with an inverted L network is shown, in accordance with an embodiment. The impedance match 430 may comprise a main line 434 and an arm 433. The main line 434 may comprise a set passive elements such as inductors, capacitors, and resistors, and the arm 433 may comprise of passive elements such as a capacitor or an inductor that are coupled to a ground. In an embodiment, an input 431 of the impedance match 430 is fed RF power from a power splitter (not shown), and the output 432 provides RF power to a chamber (not shown).

[0059] Referring now to FIG. 4C, a circuit diagram of an impedance match 430 with a two stage L network is shown, in accordance with an embodiment. The impedance match 430 may comprise a main line 434 and a pair of arms 433A and 433B. The main line 434 may comprise a set of inductors with a capacitor, and the arms 433A and 433B may each comprise a capacitor and an inductor that are coupled to a ground. In an embodiment, an input 431 of the impedance match 430 is fed RF power from a power splitter (not shown), and the output 432 provides RF power to a chamber (not shown).

[0060] Referring now to FIG. 5, a schematic illustration of an RF plasma processing tool 500 is shown, in accordance with an embodiment. In an embodiment, the RF plasma processing tool 500 may be similar to any of the RF plasma processing tools described in greater detail herein. For example, the RF plasma processing tool 500 may comprises an RF power distribution system with an RF generator 510 that feeds RF power to a power splitter 520. The power splitter 520 may split the RF power into a plurality of branches, with each branch comprising an impedance match 530 and a plasma load 545 (e.g., a plasma within a chamber (not shown)). For example, four branches with impedance matches 530A-530D with respective plasma loads (1-4) 545 are shown in FIG. 5.

[0061] In an embodiment, the power splitter 520 may be high isolation power splitter 520. Though, any power splitter 520 described herein may be used in accordance with an embodiment. For example, the power splitter 520 may comprise a plurality of lines 522A-522D with bridge 524A between lines 522A and 522B and bridge 524B between lines 522C and 522D. In an embodiment, a common node 526 electrically couples the bridge 524A to the bridge 524B.

[0062] In an embodiment, coaxial cables 528 may be used to couple the power splitter 520 to the plurality of impedance matches 530. Additionally, a sensor 529 (e.g., Sensors 1-4) may be provided

along each branch between the power splitter **520** and the impedance match **530**. The sensor **529** may be a reflected power sensor or any other sensor suitable for detecting impedance mismatches along the branch. The sensor **529** may send signals relating to impedance mismatches (e.g., reflected power percentages) to an RF power distribution system controller (not shown in FIG. 5). [0063] In an embodiment, the impedance matches **530A-530D** may have any suitable impedance matching network, such as those described in greater detail herein. In a particular embodiment, the impedance matches **530** are variable in that one or more passive components **538** are controllable by one or more motors **537** in order to change an impedance of the impedance match **530**. For example, in impedance match **530A**, motor (M1A) **537** controls passive component **538A** and motor (M1B) **537** controls passive component **538B**. The motors **537** may be driven by a motor controller **536**. The motor controller **536** is communicatively coupled to an RF power distribution system controller (not shown in FIG. 5).

[0064] Referring now to FIG. 6, a block diagram of an RF power distribution system controller **650** is shown, in accordance with an embodiment. The controller **650** may include a board **661**, such as a printed circuit board (PCB) assembly. In an embodiment, a coupling circuit and analog control circuitry **668** is provided on the board **661**. The coupling circuit and analog control circuitry **668** receives signals from the sensors **629** that are provided in the RF power distribution system (e.g., sensors **529** in FIG. 5). The signals may represent a one or more of a percentage of reflected power along each branch, an impedance mismatch measurement, transmitted power, power transmitted to the chamber, or the like. The number of sensors **629** may be equal to a number of branches in the RF power distribution system. In an embodiment, the processed signals are passed to a programmable logic block **662** that comprises an RF sensor control logic block **670**. Additionally, the processed signals are passed to a coupling circuit and dual analog/digital block **669** on the board **661**. The block **669** passes the signals to a digital down conversion (DDC) block **671** and into a frequency measuring block **673**, a transmission line scaling block **672**, and a dynamic memory access (DMA) **674**.

[0065] In an embodiment, the RF sensor control logic block **670** and the transmission line scaling block **672** may provide signals to a real time processing unit (RTPU) **663**. In an embodiment, the RTPU **663** may comprise a plurality of different processing blocks **678-683** for determining how to modify the impedance matches in order to reduce or eliminate reflected power in the RF power distribution system. In the illustration, each block **678-683** comprises multiple instances (e.g., three instances are shown). More generally, the number of instances of each block **678-683** may be equal to the number of branches in the RF power distribution system.

[0066] In an embodiment, block **678** may be an impedance match block configured to monitor the settings of the various impedance matches (not shown) in the RF power distribution system. The block **679** may be an impedance match calibration agent used to calibrate the individual impedance matches. Block **680** may be an impedance match frequency agent that is used to monitor and/or control a frequency within each impedance match. The power model control **681** may be used to control a model of the power that passes through the RF power distribution system. The safe operating area (SOA) supervisor control **682** may include a set of parameters that limit the operation of the RF plasma processing tool outside of the SOA. This can be used in order to protect the RF power distribution system from damage due to excessive reflected power and/or operating outside of set safety limits. The sensor calibration agent **683** may be used to implement calibration of the plurality of sensors **629**. In an embodiment, each of the blocks **678-683** may be electrically coupled to a memory **675**, such as a DRAM. In an embodiment, the RTPU may also be electrically coupled to a host processor **665** that provides high level control of the controller **650**.

[0067] In an embodiment, the memory **675** may be electrically coupled to a message controller **664**. The message controller **664** may be a component that manages the flow of data between blocks and/or systems within the controller **650**. For example, the message controller **664** may provide data to an interface **684** (e.g., an ENET interface, a USB interface, etc.), an ECAT module

685, and motor controller interfaces **686**. In an embodiment, the motor controller interfaces **686** may be communicatively coupled to the motor controllers **536** (shown in FIG. 5) of the impedance match. The number of motor controller interfaces **686** may match the number of impedance matches in the RF power distribution system. For example, motor controller interfaces **686A** include a pair of motor controllers (1, 2), and motor controller interfaces **686B** include a pair of motor controllers (3, 4). However, each motor controller interface may be responsible for communicating with one or more motor controllers **536**. That is, each motor controller interface **686** is not restricted to controlling a single match. That is, the RF power distribution system controller **650** may receive data from the sensors **629**, process that data to determine how the impedance matches need to be adjusted to reduce reflected power, and use the motor controllers **686** to initiate a change in the impedance matches.

[0068] In an embodiment, the programmable logic **662**, the RTPU **663**, the message controller **664**, and the host processor **665** may be integrated within a larger RF control platform **666** that is provided on the board **661**. The RF control platform **666** may be implemented on its own board that is mounted to the board **661** in some embodiments.

[0069] In an embodiment, the RF power distribution system controller **650** provides multiple benefits when used in conjunction with the hardware of the RF power distribution systems described in greater detail herein. For example, the controller **650** allows for improved RF power balancing along each of the branches in the RF power distribution system. In a particular embodiment, coupling circuitry and analog control circuitry **668** is configured to receive incident and reflected RF voltages and RF current signals from a plurality of sensors **629**. This information can then be processed through one or more different modules and/or blocks within the controller **650** in order to output a control effort that is sent to the impedance matches by the motor controller interfaces **686**.

[0070] Different blocks are described with respect to the controller **650**. The individual blocks may be implemented through hardware (e.g., circuitry), software, firmware, or any combination of the three. Connections between blocks (indicated by arrows) may represent electrical interconnects (e.g., comprising one or more of electrically conductive traces, vias, or pads) that are provide on the board **661** of any other board or package used to assemble the controller **650**.

[0071] Referring now to FIG. 7, a block diagram of an exemplary computer system **700** of a processing tool is illustrated in accordance with an embodiment. In an embodiment, computer system **700** is coupled to and controls processing in the processing tool. Computer system **700** may be connected (e.g., networked) to other machines in a Local Area Network (LAN), an intranet, an extranet, or the Internet. Computer system **700** may operate in the capacity of a server or a client machine in a client-server network environment, or as a peer machine in a peer-to-peer (or distributed) network environment. Computer system **700** may be a personal computer (PC), a tablet PC, a set-top box (STB), a Personal Digital Assistant (PDA), a cellular telephone, a web appliance, a server, a network router, switch or bridge, or any machine capable of executing a set of instructions (sequential or otherwise) that specify actions to be taken by that machine. Further, while only a single machine is illustrated for computer system **700**, the term “machine” shall also be taken to include any collection of machines (e.g., computers) that individually or jointly execute a set (or multiple sets) of instructions to perform any one or more of the methodologies described herein. In other embodiments, the term “machine” may also cover a state machine that may not have an explicit instruction set.

[0072] Computer system **700** may include a computer program product, or software **722**, having a non-transitory machine-readable medium having stored thereon instructions, which may be used to program computer system **700** (or other electronic devices) to perform a process according to embodiments. A machine-readable medium includes any mechanism for storing or transmitting information in a form readable by a machine (e.g., a computer). For example, a machine-readable (e.g., computer-readable) medium includes a machine (e.g., a computer) readable storage medium

(e.g., read only memory (“ROM”), random access memory (“RAM”), magnetic disk storage media, optical storage media, flash memory devices, etc.), a machine (e.g., computer) readable transmission medium (electrical, optical, acoustical or other form of propagated signals (e.g., infrared signals, digital signals, etc.)), etc.

[0073] In an embodiment, computer system **700** includes a system processor **702**, a main memory **704** (e.g., read-only memory (ROM), flash memory, dynamic random access memory (DRAM) such as synchronous DRAM (SDRAM) or Rambus DRAM (RDRAM), etc.), a static memory **706** (e.g., flash memory, static random access memory (SRAM), etc.), and a secondary memory **718** (e.g., a data storage device), which communicate with each other via a bus **730**.

[0074] System processor **702** represents one or more general-purpose processing devices such as a microsystem processor, central processing unit, or the like. More particularly, the system processor may be a complex instruction set computing (CISC) microsystem processor, reduced instruction set computing (RISC) microsystem processor, very long instruction word (VLIW) microsystem processor, a system processor implementing other instruction sets, system processors implementing a combination of instruction sets, or a state machine that may not have an explicit instruction set. System processor **702** may also be one or more special-purpose processing devices such as an application specific integrated circuit (ASIC), a field programmable gate array (FPGA), a digital signal system processor (DSP), network system processor, or the like. System processor **702** is configured to execute the processing logic **726** for performing the operations described herein.

[0075] The computer system **700** may further include a system network interface device **708** for communicating with other devices or machines. The computer system **700** may also include a video display unit **710** (e.g., a liquid crystal display (LCD), a light emitting diode display (LED), or a cathode ray tube (CRT)), an alphanumeric input device **712** (e.g., a keyboard), a cursor control device **714** (e.g., a mouse), and a signal generation device **716** (e.g., a speaker).

[0076] The secondary memory **718** may include a machine-accessible storage medium **731** (or more specifically a computer-readable storage medium) on which is stored one or more sets of instructions (e.g., software **722**) embodying any one or more of the methodologies or functions described herein. The software **722** may also reside, completely or at least partially, within the main memory **704** and/or within the system processor **702** during execution thereof by the computer system **700**, the main memory **704** and the system processor **702** also constituting machine-readable storage media. The software **722** may also be directly implemented as hardware gates in some embodiments. The software **722** may further be transmitted or received over a network **761** via the system network interface device **708**. In an embodiment, the network interface device **708** may operate using RF coupling, optical coupling, acoustic coupling, or inductive coupling.

[0077] While the machine-accessible storage medium **731** is shown in an exemplary embodiment to be a single medium, the term “machine-readable storage medium” should be taken to include a single medium or multiple media (e.g., a centralized or distributed database, and/or associated caches and servers) that store the one or more sets of instructions. The term “machine-readable storage medium” shall also be taken to include any medium that is capable of storing or encoding a set of instructions for execution by the machine and that cause the machine to perform any one or more of the methodologies. The term “machine-readable storage medium” shall accordingly be taken to include, but not be limited to, solid-state memories, and optical and magnetic media.

[0078] In the foregoing specification, specific exemplary embodiments have been described. It will be evident that various modifications may be made thereto without departing from the scope of the following claims. The specification and drawings are, accordingly, to be regarded in an illustrative sense rather than a restrictive sense.

Claims

- 1.** An apparatus, comprising: a radio frequency (RF) power supply; a power splitter electrically coupled to the RF power supply; a plurality of impedance matches electrically coupled to the power splitter, wherein the power splitter is configured to split RF power from the RF power supply in order to supply the RF power to each of the plurality of impedance matches; and a plurality of chambers, wherein each chamber is electrically coupled to a different one of the plurality of impedance matches.
- 2.** The apparatus of claim 1, wherein a number of the plurality of chambers is equal to a number of the plurality of impedance matches.
- 3.** The apparatus of claim 1, wherein a number of the plurality of impedance matches is greater than a number of the plurality of chambers.
- 4.** The apparatus of claim 1, wherein at least one of the plurality of chambers is configured to process a plurality of substrates in a batch process.
- 5.** The apparatus of claim 1, wherein the power splitter is a high isolation power splitter.
- 6.** The apparatus of claim 1, wherein the power splitter is a low isolation power splitter.
- 7.** The apparatus of claim 1, wherein the plurality of impedance matches comprise a pi network match, an L network match, an inverted L network match, or a two stage L network match.
- 8.** The apparatus of claim 1, wherein each of the plurality of impedance matches is independently controllable.
- 9.** The apparatus of claim 1, further comprising: a plurality of reflected power sensors, wherein each of the plurality of reflected power sensors is electrically coupled between the power splitter and a different one of the plurality of impedance matches.
- 10.** The apparatus of claim 1, further comprising: a process controller communicatively coupled to the plurality of impedance matches configured to control impedances of each of the plurality of impedance matches.
- 11.** An apparatus, comprising: a radio frequency (RF) power generator; a power splitter coupled to the RF power generator; a plurality of impedance matches, wherein each of the plurality of impedance matches is coupled to the RF power generator, and wherein the power splitter is configured to split RF power from the RF power generator to supply RF power to each of the plurality of impedance matches, and wherein each of the plurality of impedance matches comprises an actuator configured to alter an impedance of a respective impedance match; a plurality of reflected power sensors, wherein each of the plurality of reflected power sensors is electrically coupled between the power splitter and a different one of the plurality of impedance matches; and a controller, wherein the controller is configured to receive an input signal from each of the plurality of reflected power sensors and use the input signal to drive the actuators in order to alter the impedance of the respective impedance match.
- 12.** The apparatus of claim 11, wherein each of the plurality of impedance matches comprises two actuators.
- 13.** The apparatus of claim 11, wherein the controller comprises a real time processing unit (RTPU).
- 14.** The apparatus of claim 13, wherein the RTPU comprises one or more of: an impedance match module, an impedance match calibration agent, and an impedance match frequency agent.
- 15.** The apparatus of claim 13, wherein the RTPU comprises a safe operating area (SOA) supervisor control.
- 16.** The apparatus of claim 11, wherein the plurality of impedance matches comprise a pi network match, an L network match, an inverted L network match, a multi-stage L network match, or a multi-stage inverted L network match.
- 17.** The apparatus of claim 11, wherein the power splitter is a high isolation power splitter.
- 18.** An apparatus for controlling a plurality of impedance matches, comprising: coupling circuitry and analog control circuitry configured to receive incident and reflected radio frequency (RF)

voltages and RF current signals from a plurality of sensors; an RF control platform, wherein the RF control platform comprises: programmable logic; a real time processing unit (RTPU); a message controller; and a processor; and a plurality of motor control interfaces, wherein the plurality of motor control interfaces are configured to control impedances of the plurality of impedance matches.

19. The apparatus of claim 18, wherein the RTPU comprises one or more of: an impedance match module, an impedance match calibration agent, and an impedance match frequency agent, a safe operating area (SOA) supervisor control, a power model control, or a sensor calibration agent.

20. The apparatus of claim 18, wherein the apparatus is configured to provide impedance tuning and/or RF power balancing.
